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I'm a Software Engineer with an MSc (Distinction) in Software Engineering from the University of Westminster, and I've published research on AI-driven real-time navigation systems. My background is actually in Electronics & Communication Engineering, so alongside software I'm comfortable with power electronics, circuitry, microcontrollers, and IoT, which shapes how I think about systems end to end, not just the software layer. Over time I've moved deeper into machine learning, full-stack development, and building systems that scale, working on things like recommendation engines, graph-based routing algorithms, and cutting VRAM requirements by 42.9% for a production diffusion-based virtual try-on pipeline. I've also built automation tools and led multidisciplinary teams. What I care about long-term is research at the intersection of AI, intelligent decision-making, and human-centred applications, work that has a real footprint, whether that's societal or industrial.

RESEARCH INTEREST

- Machine learning and AI applied to real time, high stakes decision systems, including energy management, autonomous systems, and intelligent recommendation, where accuracy, speed, and reliability directly affect outcomes.
- AI based forecasting and optimisation methods for renewable energy management, smart grid resilience, and battery storage, supporting improved efficiency and the transition to low carbon systems.
- Trustworthy, verifiable AI assisted software development, with an emphasis on explainability, testability, and reproducibility as AI becomes increasingly embedded in the software development process.
- Distributed, sensor driven intelligent systems, combining machine learning with embedded and IoT architectures to support adaptive, real time decision making.
- Computer vision and deep learning for perception in autonomous and safety critical systems, building on published research in AI based navigation.

ACADEMIC CREDENTIALS

MSc in Software Engineering, University of Westminster, United Kingdom, **(Distinction)**, 2024.

BSc in Electronics and Communication Engineering, National University Bangladesh **(1st Class: CGPA 3.48 out of 4.00)**, 2019.

RESEARCH PUBLICATION

Hasan, M. Islam F. and Sultana, T. (2024). Real-Time Fire Navigation System Using Computer Vision and AI. International Journal of Scientific and Research Publications, [online] 14(6), p.182. Available at: <https://www.ijsrp.org/research-paper-0624/ijsrp-p15016.pdf>

RELEVANT PROJECTS

AI Virtual Try-On for Fashion E-Commerce (Cygnowear Ltd): [Project GitHub Link](#)

- Built a diffusion-based (SDXL/IDM-VTON) virtual try-on pipeline for a denim e-commerce platform, reducing peak inference VRAM by 42.9% (16.28 GB → 9.30 GB) through custom, from-scratch INT8 and INT4 weight quantisation (no third-party quantisation library).
- Benchmarked five configurations end-to-end (VRAM, latency, colour accuracy) and identified that INT8/INT4 preserve output colour while attention slicing and xFormers+upcast both cause colour drift, tracing the root cause to float16 operation-order non-associativity and a CUDA-kernel boundary issue with upcast_attention.
- Halved the minimum production GPU tier (24 GB → 12 GB) and cloud cost (~\$1.00/hr A10G → ~\$0.50/hr T4) by shipping INT8 as the default configuration, validated across 13 catalogue products.
- Designed the production architecture (FastAPI + Celery/Redis job queue + Docker, GPU worker with concurrency=1) and a batch-testing engine that cut redundant model reloads from 52 to 4 across product/config combinations.

AI Based Ecommerce Platform: [Project GitHub Link](#)

- Built a full-stack platform using ASP.NET Core, Entity Framework & Identity, PostgreSQL, React + TypeScript + Vite, and Flask/Python for ML services.
- Trained and evaluated Random Forest, KNN, SVM, and K-Means on transactional data; selected Random Forest based on accuracy, precision, recall, F1-score, and confusion matrices.
- Developed a Flask-based recommendation API integrated with frontend/backend for real-time product suggestions.
- Applied MVC architecture, UML diagrams, ER models, and OOP principles (abstraction, encapsulation, modularity) for scalable, modular design.
- Conducted unit, integration, and black-box testing of ML endpoints and ensured performance, security, and GDPR compliance for HTTPS, and authentication tokens.

PROFESSIONAL EXPERIENCE

Cygnowear Ltd, London, UK

AI Engineer

November, 2025- Present

- Built a virtual try-on feature using diffusion models (SDXL/IDM-VTON), cutting GPU memory use nearly in half via custom INT8/INT4 quantisation built from scratch.
- Benchmarked five configurations for speed, memory, and quality, traced a colour-accuracy bug to a low-level GPU floating-point issue, and shipped the setup that kept quality high at a fraction of the memory cost.
- Halved cloud hosting costs by cutting production GPU requirements, validated across the full product catalogue.
- Designed the production backend (API, job queue, containerised GPU workers) and built a testing tool that cut redundant model reloads by over 90%.
- Fine-tuned and evaluated the model against real catalogue images to improve output quality on the business's actual products.
- Partnered with product and design to define how the feature should look and behave, translating requirements into model behaviour.
- Monitored the model in production and retrained or adjusted it as new products and edge cases emerged.

Cygnoverse Ltd, London, UK

Software Engineer

July, 2025 – November, 2025

- Led a team of 8 software engineers in developing scalable software solutions, contributing to architecture design, algorithm development, and system optimisation.
- Conducted data-driven experimentation and feasibility analysis to guide product decisions, applying machine learning and systems engineering principles.
- Collaborated cross-functionally in a research-style environment, documenting findings and presenting technical insights to stakeholders.

TECHNICAL SKILLS

Research & Tools: Zotero, Mendeley, STATA, LaTeX; MATLAB, SOLIDWORKS, VS Code, NI Multisim, Proteus

Core ML/DL: Machine Learning, Deep Learning, Supervised & Unsupervised Learning, Regression, Classification, Random Forest, KNN, SVM, Decision Trees, K-Means Clustering, Convolutional Neural Networks (CNN)

Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV

Generative AI & Agentic Systems: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI, Multimodal Foundation Models, Vision-Language-Action Models, LangChain, Hugging Face Transformers, Vector Databases, Embeddings

Trustworthy & Explainable AI: Explainable AI (XAI), Uncertainty Estimation, Model Evaluation (Accuracy, Precision, Recall, F1-Score, Confusion Matrix, ROC-AUC, Jaccard Index, Log Loss), AI Safety & Verification

Distributed & Applied AI: Federated Learning, Graph Neural Networks (GNN), Physics-Informed Machine Learning, Digital Twins, Reinforcement Learning

Programming & Frameworks: C#, C, C++, Python, JavaScript, TypeScript, Flutter, SwiftUI, Flask, ASP.NET Core, Entity Framework Core, Identity Framework, React, Axios, RESTful APIs, Swagger, Postman, NUnit, xUnit

Databases: MS SQL, PostgreSQL, SQLite

DevOps & Software Design: GitHub, Azure, AWS, Docker, Kubernetes, CI/CD, MVC, MVVM, Agile

Hardware & Embedded Systems: Raspberry Pi, Arduino, Microcontrollers, Circuit Design, IoT, Network Configuration

STANDARDIZED TEST SCORE

Graduate Record Examination (GRE): Quant (156) + Verbal (150) + AWA (3.5) Total: 309.5.

International English Language Testing System (IELTS): CEFR- C1(Proficient), Band Score overall: 7.5(Expired in 2024).

LEADERSHIP AND VOLUNTEERING

Chairperson | IEEE ISTT Student Branch (IEEE Membership 4442091)

Jan 2019-Nov 2020

- Increased membership by 135% (17 to 40 members) and grew branch funds to 26,000 BDT through strategic initiatives.
- Managed budgets and logistics for 23 events, ensuring smooth execution and impactful programming.
- Built and maintained relationships with 15 sponsors and stakeholders, boosting financial support by ~40%.

PROFESSIONAL CERTIFICATION

Certified AI Engineering Professional, International Business Machines (IBM). [Credential Link](#)

National Certified Web Developer (**Level 2**), Bangladesh Technical Education Board (BTEB) [Credential Link](#)

MikroTik Certified Network Associate (**MTCNA**), MikroTik [Credential Link](#)

Recommendations: Available on Request